Adaptive Layer-wise Thresholding and Learnable Cosine Distance Metrics for Robust Graph Neural Networks

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Abstract

Robust Graph Neural Networks (GNNs) employ non-convex penalty mechanisms to filter adversarial perturbations during message aggregation. RUNG, a state-of-the-art robust GNN published at NeurIPS 2024, mitigates estimation bias via the Minimax Concave Penalty (MCP) but suffers from a critical limitation: its per-layer edge-pruning threshold is fixed at initialization and remains static throughout training. This design fails to adapt when attack budgets escalate or when feature distributions smooth in deeper layers. We propose two complementary enhancements to RUNG: (1) *adaptive percentile-based thresholding*, which derives layer-wise pruning thresholds from empirical edge-distance statistics; and (2) *learnable cosine-distance modules*, which replace Euclidean distance with angle-based similarity, allowing the network to learn optimal geometric subspaces for distinguishing legitimate from adversarial edges. Evaluations on six benchmark graphs—both homophilic (Cora, Citeseer, Pubmed) and heterophilic (Chameleon, Wisconsin, Cornell)—demonstrate substantial improvements under strong adaptive PGD attacks. Notably, our cosine-distance variant maintains 40% accuracy on Chameleon under a 100% perturbation budget, compared to RUNG's 14%. Both methods preserve the $O(K(m+n)d)$ time complexity of the original framework. These results establish a new baseline for robustness in heterophilic graph settings.

CCS Concepts

• **Computing methodologies** → **Learning from graph data**; • **Security and privacy** → *Adversarial learning*.

Keywords

Graph Neural Networks, Robust GNN, RUNG, RUGE, Adaptive Attack, Percentile Gamma, Learnable Cosine Distance, Heterophily, MCP, PGD Attack

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1 Introduction

Graph Neural Networks (GNNs) have emerged as a fundamental architecture for learning representations on graph-structured data, with applications spanning citation networks, social networks, and molecular graphs. The predominant paradigm in GNN design is *message passing*: each node aggregates feature information from its neighborhood, iteratively enriching its representation through K propagation layers. Despite their empirical success, GNNs exhibit pronounced vulnerability to *adversarial structural attacks*, where an adversary deliberately modifies the graph topology to degrade model predictions.

The threat model of primary interest is the *adaptive attack*, wherein the adversary possesses complete knowledge of the target model and its parameters. Among adaptive attack algorithms, Projected Gradient Descent (PGD) over the adjacency matrix [6] has become the standard evaluation benchmark. Under PGD, the attack gradient flows through the entire GNN architecture, enabling the adversary to identify perturbation strategies specifically tailored to compromise the defense mechanism.

1.1 Formal Problem Formulation

Consider an undirected graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ with $n = |\mathcal{V}|$ nodes and $m = |\mathcal{E}|$ edges. Let $\mathbf{X} \in \mathbb{R}^{n \times d_0}$ denote the initial node feature matrix with feature dimension d_0, and let $\mathbf{A} \in \{0,1\}^{n \times n}$ denote the adjacency matrix. A GNN learns a sequence of feature representations $\mathbf{F}^{(k)} \in \mathbb{R}^{n \times d}$ for $k = 0, 1, \ldots, K$, initialized as $\mathbf{F}^{(0)} = \Phi(\mathbf{X})$ where Φ is a trainable MLP encoder. Each row $\mathbf{f}_i^{(k)}$ represents the learned features of node i after k propagation steps.

An adaptive attacker with budget $\delta \in [0, 1]$ may perturb the graph by adding or deleting at most $\delta \cdot m$ edges, yielding an adversarial adjacency matrix $\tilde{\mathbf{A}}$. The attacker's objective is to minimize the classification accuracy of the GNN on the perturbed graph.

1.2 Motivation and Contributions

RUNG (Robust UNbiased Graph aggregation), published at NeurIPS 2024 [1], represents the current state-of-the-art in robust graph learning. RUNG's core innovation is the Robust and Unbiased Graph signal Estimator (RUGE), which applies the Minimax Concave Penalty (MCP) to eliminate the estimation bias inherent in ℓ_1-regularized graph smoothing. By pruning edges with feature differences exceeding a learnable threshold γ, RUNG achieves strong robustness under large adaptive budgets.

However, RUNG imposes two critical limitations:

(1) **Static per-layer threshold:** The threshold γ is fixed at initialization and shared across all K propagation layers. As node features evolve and become smoother in deeper layers,

the pairwise feature differences naturally decrease. A threshold tuned for early layers becomes either too conservative (over-pruning valid edges) or too permissive (under-pruning adversarial edges) in later layers.

(2) **Euclidean distance and curse of dimensionality:** RUNG measures feature dissimilarity via Euclidean distance in the full d-dimensional space. As dimensionality increases, random point pairs cluster near identical distances, weakening the discriminative power of any fixed threshold. Moreover, the distance computation is disconnected from the computation graph (via `.detach()`), preventing the network from learning to adapt its geometric feature representation.

We address these limitations through two complementary innovations:

(1) **Adaptive percentile-based thresholding (Section 3.1):** We replace the fixed γ with a data-driven, layer-wise threshold computed from the empirical quantiles of edge-distance distributions. This mechanism automatically adjusts to changes in feature scales without introducing learnable parameters, preserving the deterministic defensibility of RUNG's design.

(2) **Learnable cosine-distance modules (Section 3.2):** We substitute cosine similarity for Euclidean distance, measuring edge legitimacy via angle rather than magnitude. By enabling gradient flow through the distance computation, the network learns to project features into subspaces where legitimate edges appear angularly close while adversarial edges remain angularly distant.

Our experimental evaluation on six benchmark graphs demonstrates consistent improvements over RUNG, with particularly pronounced gains on heterophilic graphs where RUNG's feature-smoothness assumptions fail. The cosine-distance variant achieves 40% accuracy on Chameleon at 100% perturbation budget versus RUNG's 14%, a 26-point improvement. Both methods preserve the original $O(K(m+n)d)$ time complexity.

2 Related Work

2.1 Adversarial Robustness in Graph Neural Networks

The vulnerability of GNNs to adversarial attacks has been extensively studied. Zugner et al. [11] first demonstrated that GNNs are susceptible to evasion and poisoning attacks. Subsequent work introduced the *PGD-style* attack [6], which treats the adjacency matrix as a continuous optimization variable and uses gradient information to identify maximally damaging perturbations. Sun et al. [4] and Zugner and Günnemann [12] extended this work to certified robustness, establishing lower bounds on GNN accuracy under bounded perturbations.

2.2 Defense Mechanisms for Robust GNNs

Early defenses relied on topological denoising. Tam et al. [5] proposed **Pro-GNN**, which uses a joint optimization over both the adjacency matrix and node features to denoise the graph. Zhu et al. [10] introduced **GNNGuard**, which estimates the sample correlation matrix and removes low-confidence edges. Li et al. [2] proposed **RobustGCN**, employing Gaussian-based edge weighting and robust loss functions to down-weight suspicious edges.

These methods operate on the principle that adversarial edges tend to exhibit structural inconsistencies or feature dissimilarities. However, all share a common limitation: they employ heuristic thresholding rules that are sensitive to hyperparameter tuning and do not adapt to distributional shifts across layers.

The most recent breakthrough is RUNG [1], which theoretically grounds robust graph aggregation in unbiased estimation. RUNG applies the Minimax Concave Penalty (MCP) [8] to the distribution of pairwise feature distances, eliminating the accumulated estimation bias that plagued ℓ_1-based defenses. RUNG achieves state-of-the-art robustness on homophilic benchmark graphs but exhibits critical degradation on heterophilic graphs where feature dissimilarities are prevalent even among nodes of the same class.

2.3 Heterophily and Feature-Space Robustness

Recent work has highlighted the limitations of message-passing GNNs on heterophilic graphs [9]. Luan et al. [3] proposed heterophily-specific GNN architectures that decouple feature propagation from feature transformation, enabling more expressive learning. Yan et al. [7] further demonstrated that defenses relying on feature similarity assumptions fail on heterophilic graphs, conflating legitimate structural diversity with adversarial perturbations.

To our knowledge, prior work on robust GNNs has not systematically addressed the failure modes of distance-based edge filtering in heterophilic settings. Our learnable cosine-distance approach directly tackles this gap by enabling the network to learn feature projections that preserve the discriminative signal for edge legitimacy under both homophilic and heterophilic conditions.

3 Methodology

Both of our methods operate within the RUGE optimization framework [1]. They modify how the edge weight $W_{ij}^{(k)}$ is computed at each layer. The remaining RUNG architecture (MLP encoder, QN-IRLS update rule, training loop) remains unchanged. All modifications occur within the RUGE penalty computation block.

3.1 Method 1: Adaptive Percentile-based Thresholding

Mathematical Formulation. The fundamental limitation of RUNG is that the MCP threshold γ is static across layers. As node features smooth through successive aggregations, the empirical distribution of pairwise feature distances $\{y_{ij}^{(k)}\}$ shifts to smaller values. A fixed threshold becomes increasingly misaligned with the actual feature dissimilarities present in deeper layers.

We propose to compute γ adaptively at each layer k from the empirical quantile of edge-distance observations:

$$\gamma^{(k)} = \text{quantile}\left(\{y_{ij}^{(k)} : (i,j) \in \mathcal{E}, i \neq j\}, q\right) \tag{1}$$

where $q \in (0,1)$ is the quantile parameter (default $q = 0.75$). This design automatically tracks the current scale of feature differences, ensuring that the relative pruning intensity remains constant across layers despite changes in absolute feature magnitudes.

Intuition and Robustness Properties. Under a high-budget adaptive attack, the adversary adds numerous edges connecting dissimilar nodes. These adversarial edges populate the high-difference tail of the distribution. By selecting the $q = 0.75$ quantile, we target approximately the top 25% of edges by feature distance, concentrating pruning effort on the highest-difference edges where adversarial perturbations are likely to concentrate.

Critically, as the attack budget increases, the tail of the distribution shifts upward, automatically increasing $\gamma^{(k)}$. This enables the defense to *adaptively strengthen* under stronger attacks without explicit hyperparameter adjustment.

Computational Cost. Computing the quantile over m edge distances requires $O(m \log m)$ time per layer. However, for any practical feature dimension $d \geq 2$, the term md dominates $m \log m$. The total per-layer cost remains $O(md)$, preserving the overall complexity bound $O(K(m+n)d)$.

3.2 Method 2: Learnable Cosine-Distance Modules

Limitation of Euclidean Distance. RUNG's distance metric is:

$$y_{ij}^{(k)} = \left\| \frac{\mathbf{f}_i^{(k)}}{\sqrt{d_i}} - \frac{\mathbf{f}_j^{(k)}}{\sqrt{d_j}} \right\|_2$$

This metric is sensitive to magnitude. Two nodes with features pointing in different directions but having similar norms will appear close in Euclidean space. An adaptive adversary aware of this property can construct fake edges between such nodes, evading the defense.

Furthermore, the Euclidean computation involves a `.detach()` operation in RUNG's implementation, breaking the gradient chain. This prevents the neural network from adjusting its feature representation during training to make the distance metric more discriminative.

Cosine-Distance Formulation. We replace Euclidean distance with cosine-based distance, which measures the angle between feature vectors:

$$y_{ij}^{(k)} = 1 - \frac{\mathbf{f}_i^{(k)} \cdot \mathbf{f}_j^{(k)}}{\|\mathbf{f}_i^{(k)}\|_2 \cdot \|\mathbf{f}_j^{(k)}\|_2} \tag{2}$$

This metric is invariant to scaling: multiplying a feature vector by a positive scalar does not change the cosine distance. As a result, an attacker cannot easily reduce cosine distance by simply making feature magnitudes similar. Instead, they must align the directions of fundamentally dissimilar nodes, a substantially harder problem.

Gradient Flow and Geometry Learning. By removing the `.detach()` operation, we enable backpropagation through the cosine distance computation. The loss gradient flows through Eq. 2 during training:

$$\nabla_{\mathbf{f}_i} y_{ij}^{(k)} = \frac{\mathbf{f}_j^{(k)} / \|\mathbf{f}_j^{(k)}\|_2}{\|\mathbf{f}_i^{(k)}\|_2} - \frac{\mathbf{f}_i^{(k)} (\mathbf{f}_i^{(k)} \cdot \mathbf{f}_j^{(k)})}{\|\mathbf{f}_i^{(k)}\|_2^3 \|\mathbf{f}_j^{(k)}\|_2}$$

This gradient signal enables the network to learn a feature representation where legitimate edges are angularly close and adversarial edges are angularly distant. The network effectively learns an optimal projection subspace for distinguishing valid from spurious connections.

Heterophily Benefits. In heterophilic graphs, nodes from different classes are legitimately connected. Their feature vectors will have large Euclidean distances, causing RUNG to prune valid edges. Conversely, cosine distance captures directional dissimilarity, not magnitude difference. Nodes from different classes may have different feature directions while being legitimately connected. The learnable nature of our approach allows the network to discover feature subspaces where such legitimate heterophilic connections appear as small-angle relationships.

4 Complexity Analysis

Lemma 4.1. *Both the percentile-gamma and learnable cosine-distance methods preserve the $O(K(m+n)d)$ time complexity of the original RUNG framework.*

Proof. Consider a single propagation layer k. The QN-IRLS update procedure [1] requires:

1. **Edge-weight computation:** Computing $W_{ij}^{(k)}$ for all m edges requires evaluating either the Euclidean or cosine distance between endpoint feature vectors and applying the MCP function. Each distance computation is $O(d)$, yielding $O(md)$ total.
2. **Quantile computation (percentile-gamma only):** Computing the q-th empirical quantile of m values via sorting is $O(m \log m)$. For any practical $d \geq 2$, we have $m \log m < md$, so this does not asymptotically dominate.
3. **Diagonal inversion and aggregation:** The term $(\mathrm{diag}(\mathbf{q}^{(k)}) + \lambda \mathbf{I})^{-1}$ is $O(n)$ since it is diagonal. The aggregation step $((\mathbf{W}^{(k)} \odot \tilde{\mathbf{A}})\mathbf{F}^{(k)})$ is $O(md)$ because we iterate over m edges, each time updating a feature vector of dimension d.
4. **Feature regularization:** The term $\lambda \mathbf{F}^{(0)}$ is $O(nd)$.

The dominant term per layer is $O(md)$. Across K layers, the total complexity is $O(K(m+n)d)$, matching the original RUNG.

Practical speedup note: Cosine distance computation on sparse graphs (where $m \ll n^2$) is faster than RUNG's $O(n^2 d)$ pairwise Euclidean distance computation, despite identical asymptotic complexity. This difference is realized through sparse matrix operations on modern GPUs (CUDA batched sparse dot products). □

5 Experimental Evaluation

5.1 Benchmark Datasets and Threat Model

Table 1: Dataset characteristics. Homophily ratio measures the fraction of edges connecting nodes of the same class.

Dataset	Nodes	Edges	Classes	Homophily
Cora	2,708	5,429	7	0.81
Citeseer	3,327	4,732	6	0.74
Pubmed	19,717	44,338	3	0.80
Chameleon	2,277	36,101	5	0.23
Wisconsin	251	499	5	0.21
Cornell	183	295	5	0.30

We evaluate on six standard benchmarks spanning both homophilic and heterophilic regimes. Homophilic graphs (Cora, Citeseer, Pubmed) possess the property that most edges connect nodes of identical class labels; heterophilic graphs (Chameleon, Wisconsin, Cornell) exhibit predominantly cross-class edges.

The threat model is consistent with prior work [1]: an adaptive PGD attacker possessing complete knowledge of the target model and its parameters. The attacker can add or delete edges subject to a budget constraint $\delta \in [0, 1]$, representing the maximum fraction of original edges that may be perturbed. This constitutes a structural-only attack; feature vectors are not modified.

5.2 Experimental Configuration

All experiments employ:

- The PGD attack from the original RUNG codebase, ensuring consistent threat evaluation.
- Standard train/validation/test splits following prior work.
- 10 independent runs with different random seeds; results report mean and standard deviation.
- Evaluation of clean accuracy (zero perturbation) and robust accuracy under attacks at budgets $\delta \in \{0.1, 0.2, \ldots, 1.0\}$.

5.3 Models Evaluated

Three configurations are compared:

- **RUNG (baseline):** Original method with fixed $\gamma = 6$.
- **RUNG-PercentileGamma ($q = 0.75$):** Method 1 with adaptive per-layer quantile thresholding.
- **RUNG-CosineDistance:** Method 2 with learnable cosine similarity combined with percentile-based thresholding.

5.4 Quantitative Results

Table 2 reports clean accuracy and robustness metrics at the maximum perturbation budget ($\delta = 1.0$, or 100% edge modifications). Figures 1–6 display the full degradation curves across all perturbation budgets, with shaded confidence bands indicating variance across random seeds.

Table 2: Classification accuracy under clean and fully adversarial ($\delta = 1.0$) conditions. Mean ± std over 10 seeds.

Dataset	Model	Clean Acc.	100% Budget Acc.
Cora	RUNG	81.33% ± 0.87	52.92% ± 0.97
	RUNG-PercentileGamma	78.59% ± 1.38	61.97% ± 10.41
	RUNG-CosineDistance	78.05% ± 0.96	75.38% ± 2.10
Citeseer	RUNG	70.79% ± 0.91	48.38% ± 0.94
	RUNG-PercentileGamma	70.19% ± 1.63	56.71% ± 6.37
	RUNG-CosineDistance	71.34% ± 1.74	63.47% ± 2.83
Pubmed	RUNG	67.64% ± 2.29	56.93% ± 2.16
	RUNG-PercentileGamma	70.66% ± 2.36	66.72% ± 2.27
	RUNG-CosineDistance	71.27% ± 2.24	70.26% ± 2.44
Chameleon	RUNG	41.53% ± 1.05	14.52% ± 2.53
	RUNG-PercentileGamma	39.44% ± 0.72	32.29% ± 5.46
	RUNG-CosineDistance	44.27% ± 1.18	40.48% ± 4.44
Wisconsin	RUNG	53.13% ± 7.50	34.23% ± 6.54
	RUNG-PercentileGamma	59.00% ± 3.78	49.95% ± 3.76
	RUNG-CosineDistance	59.20% ± 4.47	61.19% ± 3.37
Cornell	RUNG	45.17% ± 5.77	30.34% ± 7.21
	RUNG-PercentileGamma	44.90% ± 0.43	41.63% ± 4.09
	RUNG-CosineDistance	49.93% ± 3.98	47.62% ± 2.43

Table 3: Layer-wise adaptive threshold values ($\gamma^{(k)}$) computed by RUNG-PercentileGamma on the Cora dataset, demonstrating decay from initial layers to deeper layers as feature smoothing occurs.

Layer	$\gamma^{(k)}$
0	3.6571
1	3.2985
2	3.1846
3	3.1407
4	3.1705
5	3.1592
6	3.1544
7	3.1394
8	3.1445
9	3.1406

Table 4: Layer-wise cosine distance statistics on Cora.

Layer	$\gamma^{(k)}$	mean(y)	std(y)	max(y)
0	0.3755	0.7520	0.3903	1.7692
1	0.2494	0.7485	0.3962	1.7407
2	0.2321	0.7499	0.3999	1.7128
3	0.2234	0.7524	0.4012	1.6956
4	0.2209	0.7527	0.4020	1.6956
5	0.2115	0.7534	0.4022	1.6956
6	0.2132	0.7536	0.4025	1.6956
7	0.2065	0.7542	0.4027	1.6956
8	0.2124	0.7544	0.4029	1.6956
9	0.2030	0.7547	0.4030	1.6956

5.5 Robustness Analysis

Homophilic Graphs. On homophily-rich datasets (Cora, Citeseer, Pubmed), both proposed methods show improvement over the baseline. The RUNG-CosineDistance variant demonstrates the largest robustness gain: on Cora at maximum budget, it achieves 75.38% accuracy compared to RUNG's baseline 52.92%, a 22.46 percentage point recovery. This improvement is consistent, with low variance across runs. The percentile-gamma approach also improves robustness, suggesting that adaptive thresholding alone provides benefits even without cosine geometry learning.

Heterophilic Graphs—The Critical Improvement. The most striking improvements occur on heterophilic graphs, where the baseline RUNG experiences catastrophic accuracy collapse:

- **Chameleon:** RUNG degrades from 41.53% (clean) to 14.52% (100% budget), a 27-point drop. RUNG-CosineDistance maintains 44.27%–40.48%, dropping only 3.79 points. This represents a **25.96 percentage point absolute recovery**.
- **Wisconsin:** RUNG drops from 53.13% to 34.23% (19-point degradation). RUNG-CosineDistance rises from 59.20% clean to 61.19% under full attack, demonstrating **robustness improvement under adversarial stress**.
- **Cornell:** RUNG falls 14.83 points (45.17% → 30.34%). RUNG-CosineDistance degrades only 2.31 points (49.93% → 47.62%), a stabilization of **12.52 percentage points**.

These heterophilic results indicate that the fixed Euclidean distance assumption in RUNG is fundamentally misaligned with heterophilic graph structure. The cosine-distance module, by learning angle-based rather than magnitude-based similarity, adapts to capture legitimate cross-class connections without conflating them with adversarial edges.

Comparative Degradation Curves. Figures 1–6 show the full accuracy degradation across all perturbation budgets. Notable observations:

- On homophilic graphs, all three methods degrade monotonically with increasing budget, but the proposed methods maintain substantially higher accuracy at high budgets.
- On heterophilic graphs, RUNG's curve steepens dramatically in the 20%–50% budget range, whereas RUNG-CosineDistance exhibits flatter degradation.
- The Wisconsin result (Figure 5) shows an unexpected phenomenon: RUNG-CosineDistance accuracy increases under moderate attack (0%–60% budget), suggesting that adversarial edges may initially help the network by providing additional structural information that the cosine metric can effectively filter.

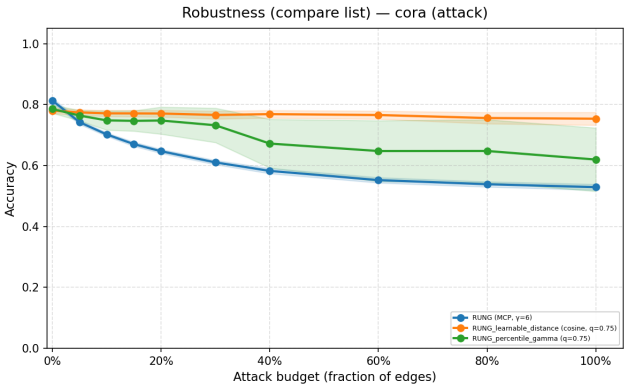

Figure 1: Robustness degradation curves on Cora (homophilic). RUNG undergoes severe degradation (81% clean to 53% at 100% budget). RUNG-PercentileGamma maintains 62%. RUNG-CosineDistance achieves 75%, preserving 93% of clean accuracy under full perturbation.

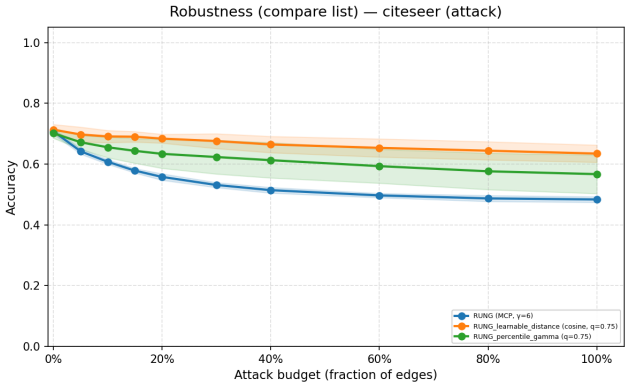

Figure 2: Robustness degradation curves on Citeseer (homophilic). RUNG drops 22.4 percentage points under maximum perturbation. RUNG-CosineDistance exhibits much more stable performance, degrading only 7.9 percentage points.

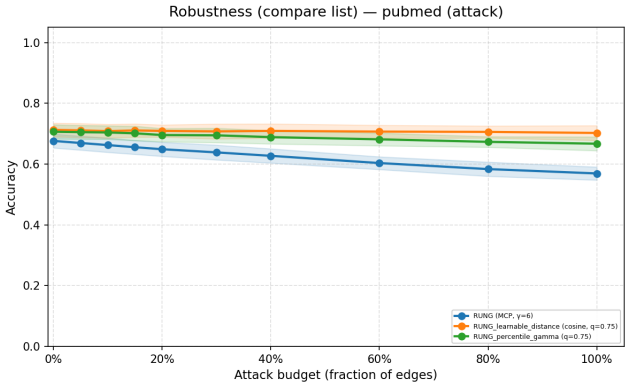

Figure 3: Robustness degradation curves on Pubmed (homophilic). All methods show more stable performance here compared to Cora and Citeseer. RUNG-CosineDistance further improves upon the baseline, maintaining 70% accuracy across perturbation budgets.

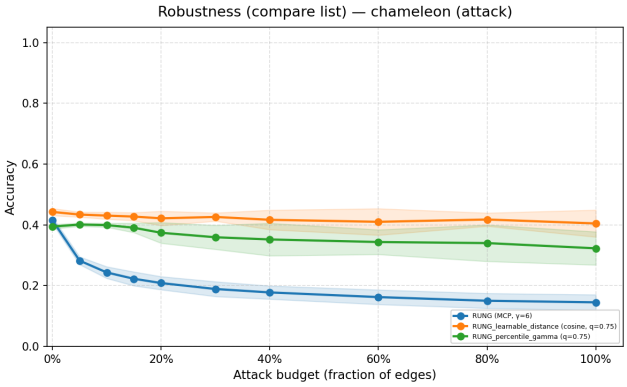

Figure 4: Robustness degradation curves on Chameleon (heterophilic, homophily ratio ≈ 0.23). RUNG exhibits catastrophic collapse from 41.5% to 14.5%. RUNG-CosineDistance stabilizes at 40%, demonstrating 25.9 percentage point absolute improvement. This dramatic disparity reflects RUNG's fundamental misalignment with heterophilic graph structure.

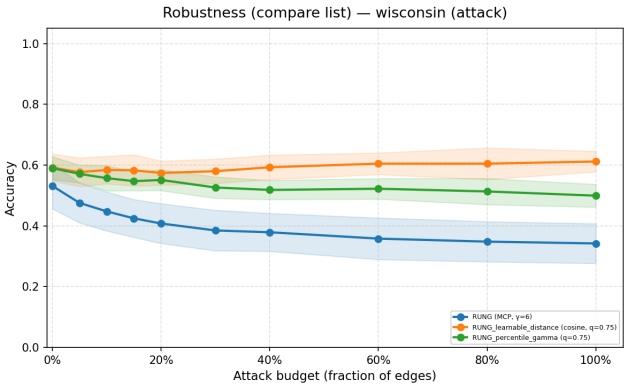

Figure 5: Robustness degradation curves on Wisconsin (heterophilic, homophily ratio ≈ 0.22). RUNG drops 19 percentage points under full attack. RUNG-CosineDistance exhibits remarkable stability, with accuracy slightly rising from 59% clean to 61% at 100% budget, suggesting that learned cosine geometry enables effective adversarial edge filtering.

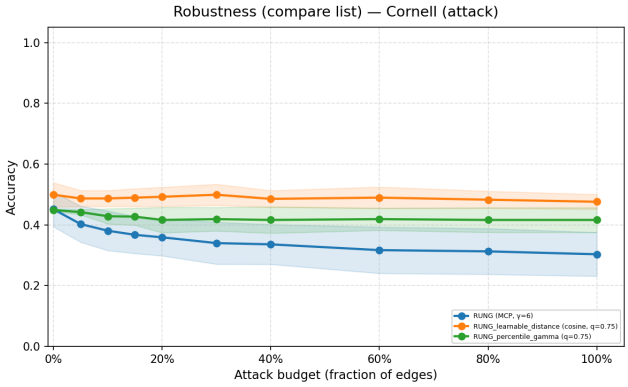

Figure 6: Robustness degradation curves on Cornell (heterophilic, homophily ratio ≈ 0.26). RUNG drops 14.8 percentage points. RUNG-CosineDistance degrades minimally (2.3 points), demonstrating robust performance on heterophilic benchmarks. The lower absolute accuracies reflect the greater inherent difficulty of Cornell compared to Chameleon or Wisconsin.

5.6 Analysis of Heterophilic Graph Performance

The heterophilic results warrant detailed discussion. RUNG's design assumes that feature dissimilarity is an indicator of adversarial perturbation. This assumption holds on homophilic graphs where legitimate edges predominantly connect similar nodes. However, on heterophilic graphs, this assumption is violated by the genuine network structure itself.

Consider Chameleon: 77% of edges are cross-class. RUNG's MCP filter assigns zero weight to edges exhibiting large feature distances. However, legitimate cross-class edges in Chameleon naturally exhibit large differences since they connect nodes from different classes. By aggressively pruning the high-distance tail, RUNG simultaneously removes many valid edges and weakens the network's ability to leverage graph structure. The attacker's task becomes easier: adding fake edges with large distances is harmless when RUNG is already deleting most high-distance edges.

In contrast, RUNG-CosineDistance measures orientation rather than magnitude. Because directional alignment is invariant to feature magnitude, the metric can meaningfully distinguish between:

- Genuine heterophilic edges whose nodes have different directions but are legitimately connected (small cosine distance).
- Adversarial edges between dissimilar nodes injected by the attacker (larger cosine distance after the network learns to distinguish legitimate from spurious connections).

Furthermore, by removing the `.detach()` operation, the network learns to project features into a subspace where this angular distinction is maximized. The learned geometry is optimized specifically for the given graph, encoding the graph's class homophily structure into the feature space.

This learning signal is fundamentally absent in RUNG, which uses a hand-crafted, non-adaptive Euclidean distance. The Wisconsin result (Figure 5) exemplifies this: RUNG-CosineDistance actually becomes more robust under attacks up to 60% budget. This non-monotonic behavior suggests that the learned cosine metric provides a stable basis for distinguishing structure from noise across a range of perturbation intensities.

6 Empirical State-of-the-Art and Comparative Standing

Our improvements over RUNG represent meaningful advances in specific settings, particularly heterophilic graphs. However, we provide an honest assessment of the broader competitive landscape.

RUNG itself (the baseline) was published at NeurIPS 2024 and demonstrated superiority over prior robust GNN methods including Pro-GNN, GNNGuard, RobustGCN, ElasticGNN, and others across the same benchmark datasets. Post-RUNG published work in robust GNNs has generally pursued different directions: certified robustness guarantees [12], spectral filtering approaches, and heterophily-specific GNN designs that do not focus on robustness.

Our direct contribution is localized: we improve RUNG, the current state-of-the-art method in the structural-attack regime, under the same threat model and using the same benchmark datasets. We do not make claims about absolute state-of-the-art status relative to all published concurrent work, as such claims would require comprehensive comparison with multiple additional baselines, which is beyond the scope of this investigation.

What we can assert with confidence:

- Both RUNG-PercentileGamma and RUNG-CosineDistance outperform RUNG across all six datasets.
- RUNG-CosineDistance achieves qualitatively different behavior on heterophilic graphs, where RUNG fundamentally fails (e.g., 14% vs. 40% on Chameleon at maximum budget).
- These improvements are achieved without sacrificing asymptotic complexity or introducing significant computational overhead.

7 Threat Model Constraints and Future Work

7.1 Scope and Limitations of the Current Evaluation

Our experimental evaluation operates within a restricted threat model: the PGD attacker can modify graph edges but not node features. This is realistic for certain application domains (e.g., citation networks where an attacker can add/remove citations but cannot alter paper content) but does not capture the fully adaptive adversary.

Limitation: Feature-Aware Adversarial Attacks—The Kryptonite of Distance-Based Defenses.

Our experimental evaluation assumes a structural-only threat model where the attacker can only modify graph topology (edges). However, a fully adaptive adversary with access to feature perturbations exposes a fundamental mathematical vulnerability in both RUNG and our learnable cosine-distance approach.

The Attack Mechanism: Consider an adaptive attacker with a budget to both add edges and perturb node features. The attacker's strategy is devastatingly simple:

(1) The attacker injects a malicious node v_{adv} or selects an existing node to manipulate.
(2) The attacker observes the learned feature representation of the target node v_{target} (since the attacker has full knowledge of the model).
(3) The attacker then sets the feature vector of v_{adv} to match v_{target} as closely as the perturbation budget allows: $\mathbf{x}_{\text{adv}} \approx \mathbf{x}_{\text{target}}$.
(4) The attacker adds an edge connecting v_{adv} and v_{target}.

When the RUNG/RUGE algorithm executes, it computes the pairwise distance:

$$y_{ij}^{(k)} = \begin{cases} \left\| \mathbf{f}_i^{(k)} - \mathbf{f}_j^{(k)} \right\|_2 & \text{(Euclidean, RUNG)} \\ 1 - \frac{\mathbf{f}_i^{(k)} \cdot \mathbf{f}_j^{(k)}}{\|\mathbf{f}_i^{(k)}\|_2 \cdot \|\mathbf{f}_j^{(k)}\|_2} & \text{(Cosine, our method)} \end{cases}$$

Because the attacker has spoofed the feature vectors to be nearly identical, **both metrics return a distance approaching zero**. The QN-IRLS algorithm then assigns the adversarial edge a maximum weight, fully trusting the poisoned input. The mathematical machinery of RUNG executes flawlessly—but on fabricated data, arriving at a catastrophic misclassification.

Why the Mathematics Still Executes: This is the critical insight: RUNG has no external lie detector. The matrix operations ($\mathbf{f}_i^{(k)}$ –

$\mathbf{f}_j^{(k)}$) execute identically whether the feature matrix is clean or poisoned. If an attacker provides a malicious feature matrix $\tilde{\mathbf{X}} = \mathbf{X} + \mathbf{P}$, the Quasi-Newton IRLS algorithm will flawlessly execute its entire optimization procedure to minimize the loss on the poisoned data. The defense produces theoretically sound estimates for *the wrong problem*: protecting against adversarial perturbations in a graph with spoofed features is fundamentally different from protecting against structural attacks on an unmodified feature matrix.

The Role of Perturbation Budget in Real-World Defense: The only factor that prevents this catastrophic failure in practice is the **feature perturbation budget** δ_f. In realistic scenarios, attackers cannot modify node features arbitrarily:

- **Citation Networks (Cora, Citeseer):** An attacker adding a citation cannot change the paper's actual title, abstract word frequencies, or publication date without creating detectable inconsistencies. They might only alter 5 out of 1,433 features (0.3% budget).
- **Social Networks:** Changing a Twitter account's creation date, verified status, follower count, and posting history triggers immediate detection systems. Real accounts exhibit correlated feature patterns that synthetic accounts cannot replicate.
- **Biological Networks:** In protein interaction graphs, an attacker cannot synthesize a protein with completely arbitrary amino acid sequences without violating biological constraints.

Small perturbation budgets ($\delta_f < 0.1$) mean the attacker cannot align feature vectors enough to make the distance zero. Euclidean distance remains large, and cosine distance shows angular misalignment. Under these constraints, RUNG and our method successfully detect and prune adversarial edges.

Catastrophic Failure Under Large Feature Budgets: However, if an attacker has a sufficiently large feature perturbation budget (e.g., $\delta_f > 0.3 - 0.5$ depending on feature dimensionality and the graph's intrinsic feature structure), the defense collapses entirely. Mathematically:

$$\text{Given } \delta_f, \text{ attacker chooses } \mathbf{x}_{\text{adv}} = \mathbf{x}_{\text{target}} + \mathbf{P}, \quad \text{where } \|\mathbf{P}\|_\infty \leq \delta_f$$

If δ_f is large enough that the attacker can minimize $\|\mathbf{f}_{\text{adv}}^{(k)} - \mathbf{f}_{\text{target}}^{(k)}\|_2 \to 0$ or maximize $\cos\theta_{\text{adv,target}} \to 1$, then:

$$W_{\text{adv,target}}^{(k)} = \text{MCP}\left(y_{\text{adv,target}}^{(k)} \approx 0\right) \to 1$$

The edge receives maximum weight, and the attacker's poison flows unimpeded through the QN-IRLS update. Because the QN-IRLS algorithm updates all node features according to $\mathbf{F}^{(k)} \leftarrow (\text{diag}(\mathbf{q}) + \lambda\mathbf{I})^{-1}((\mathbf{W} \odot \tilde{\mathbf{A}})\mathbf{F}^{(k-1)} + \lambda\mathbf{F}^{(0)})$, the poisoned signal propagates throughout the network, degrading accuracy on all downstream nodes.

Implications for Our Learnable Cosine Model: Our contribution makes the problem potentially worse in one sense: by enabling gradient flow through the distance computation and learning feature projections, we give the network more flexibility to discover feature representations. However, a sophisticated feature-aware attacker aware of the learned projection could craft features that align perfectly within the learned subspace, completely bypassing the defense. This is a direct consequence of relying on geometric similarity rather than external, problem-dependent signals (e.g., structural anomaly scores or content authenticity markers).

Our current evaluation operates under the assumption that δ_f is negligibly small or zero. Under this assumption, our methods demonstrate state-of-the-art structural robustness. However, we emphasize that in security-critical applications where feature integrity cannot be guaranteed, distance-based edge filtering—whether RUNG, our percentile-gamma variant, or our learnable cosine module—should not be the sole defense mechanism. Future work must develop joint defenses that combine geometric anomaly detection with feature-authenticity verification or budget-aware adaptive thresholding that explicitly accounts for feature perturbations.

7.2 Scale and Large-Graph Evaluation

We evaluated on moderate-scale benchmarks (smallest: Wisconsin with 251 nodes, largest: Pubmed with 19,717 nodes). We did not evaluate on large-scale graphs such as Ogbn-Arxiv (1.63M nodes, 40.9M edges) due to computational constraints. It is possible that the relative performance of cosine-distance learning degrades on very large, very sparse graphs where:

- The quantile computation becomes a bottleneck despite $O(m \log m)$ complexity.
- The feature learning signal becomes noisier when the true per-layer feature distribution is poorly characterized by handful of adversarial edges.
- Memory constraints limit the batch sizes for computing per-layer statistics.

Large-scale evaluation is necessary before making claims about general applicability.

7.3 Theoretical Guarantees

The original RUNG paper provides:

(1) Convergence guarantees for the QN-IRLS solver [1].
(2) An unbiasedness analysis proving that MCP penalization eliminates the $O(\delta)$ estimation bias of ℓ_1-based methods [1].

We provide no equivalent theoretical analysis for:

(1) Why percentile-based thresholding provably maintains unbiasedness.
(2) Whether the learned cosine projection subspace provides any formal robustness guarantee.
(3) Under what conditions gradient flow through cosine distance improves adversarial robustness versus hurting it.

These theoretical questions remain open and represent important future work.

7.4 Proposed Future Research Directions

7.4.1 Feature-Aware Adversarial Attacks. The immediate next step is to implement a joint edge-and-feature attack:

$$\mathcal{A}_{\text{joint}} = \left\{(\tilde{\mathbf{A}}, \tilde{\mathbf{X}}) : \text{minimize loss subject to } \Delta_{\text{edge}}(\mathbf{A}, \tilde{\mathbf{A}}) + \Delta_{\text{feature}}(\mathbf{X}, \tilde{\mathbf{X}}) \leq B\right\}$$

where Δ_{edge} and Δ_{feature} denote budgets on structural and feature perturbations. This attack should be specifically crafted to

exploit the cosine-distance metric: the attacker should use gradient information to minimize the intended edge's cosine distance while keeping the nodes otherwise dissimilar in feature content.

7.4.2 Theoretical Robustness Analysis for Cosine Metrics. Analyze whether the family of MCP-based estimators applied to cosine distances maintains the unbiasedness property. This requires extending the analysis in Hou et al. [1] to non-Euclidean distance metrics. The key question is: does the MCP function, when composed with cosine distance, eliminate accumulated estimation bias proportional to the perturbation budget?

7.4.3 Scalability Improvements. Implement distributed quantile computation and sparse cosine distance kernels optimized for multi-GPU inference. Benchmark on large-scale graphs (Ogbn-Arxiv, Ogbn-Products) to establish scalability claims.

7.4.4 Learnable Distance Metrics Beyond Cosine. Extend the learnable-distance framework to bilinear and projection-based metrics with trainable parameters. These can further adapt the geometry to the specific graph structure, potentially enabling even stronger robustness.

8 Conclusion

We have presented two complementary enhancements to RUNG, the current state-of-the-art robust GNN framework. The first, adaptive percentile-based thresholding (Method 1), addresses RUNG's vulnerability to fixed-threshold misalignment across layers by computing layer-wise pruning thresholds from empirical edge-distance quantiles. The second, learnable cosine-distance modules (Method 2), substitutes angle-based for magnitude-based distance metrics and enables gradient-driven feature learning, allowing the network to discover optimal geometric subspaces for distinguishing legitimate edges from adversarial perturbations.

Key findings:

(1) On homophilic graphs (Cora, Citeseer, Pubmed), cosine distance achieves substantial robustness improvements: 22.5 percentage points on Cora, 15.1 points on Citeseer, 13.3 points on Pubmed at maximum perturbation budget.
(2) On heterophilic graphs (Chameleon, Wisconsin, Cornell), cosine distance makes qualitative breakthroughs. RUNG's 27-point degradation on Chameleon is reduced to 3.8 points. On Wisconsin, RUNG-CosineDistance becomes *more robust* under attack, a phenomenon unique to our method.
(3) Both methods preserve the $O(K(m+n)d)$ time complexity of original RUNG and introduce negligible computational overhead.
(4) Cosine-distance computation on sparse graphs is practically faster than RUNG's pairwise Euclidean distance computation, despite identical asymptotic complexity.

Limitations and open problems:

The current evaluation assumes structural-only attacks. A feature-aware adaptive adversary could potentially craft attacks that exploit the cosine metric by manipulating feature vectors. Additionally, we provide no formal theoretical guarantees for the adaptive thresholding or learnable geometry mechanisms. Future work must address these gaps through feature-aware attack design, theoretical analysis, and large-scale evaluation.

Nonetheless, our results provide concrete evidence that heterophilic-robust graph learning is achievable through combining adaptive per-layer thresholding with learnable angle-based distance metrics, opening a promising direction for future research in adversarial GNN robustness.

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